Twelve slots reduce 52-bit recovery work on RTX 4090

All eight prescribed 52-bit executions recovered and independently replayed the public target scalar. On the same filtered meet-in-the-middle service, 12 slots used 251 / 261 relations, compared with 743 / 709 for 8 slots. Observed cold solve time fell from about 9.7–9.9 seconds to 2.7–3.0 seconds. The two 63-bit targets reached explicit limits with both tested arities.

Matched configurations

Seed	b: 8 slots	u: 10 slots	v: 12 slots	x: repeat v
101	9.725032	4.967727	3.032026	3.048896
202	9.889092	4.508011	2.749641	2.739248

Cold solve seconds, rounded from solver_seconds in results-v19/rows.csv. The full audit totals include fresh replay and the corrupted-scalar rejection. Order is b/u/v/x for seed101 and x/v/u/b for seed202. Same-arity repeat certificates match byte for byte; scalars agree across all configurations. Shared-host clocks are L0 with wall_valid=false; these are descriptive timings.

Seed	8 slots relations / queries	10 slots relations / queries	12 slots relations / queries
101	743 / 1050	383 / 510	251 / 341
202	709 / 1289	368 / 618	261 / 306

The 12-slot repeat has identical counts. Linear-algebra stage times were 159.268 / 141.308 ms for 8 slots and 7.643 / 11.975 ms for 12 slots. Each successful 52-bit cell used a complete right cache and a nonzero filter.

b: m8; u: m10; v/x: m12; red: timeout/limit. All points retained.

All twelve executions are shown, including the four fast limit exits. Red points are limit outcomes, not completed recoveries.

Memory and work limits, validation and custody

Seed	Slots	Admission bytes	Outcome
101	10	57,354,429,359	GPU-memory limit
101	12	64,960,046,743	GPU-memory limit
202	12	3,373,077,439	Right-side work limit
202	10	67,478,106,239	GPU-memory limit

The 12-slot seed202 split has 139,110,048 left rows and 143,721,864,000 right combinations, exceeding the 2^34 decomposition-work cap. All four 63-bit cells have zero right-cache rows. Admission bytes are conservative allocation bounds, not measured peak VRAM. No intermediate field sizes were prescribed: 53–62-bit behavior on this GPU was not measured by this panel.

One NVIDIA GeForce RTX4090, telemetry 24,564 MiB, driver595.91.07, compute capability8.9. CUDA13.2 / sm_89, Rust1.88.0. Pod25f10of8l9nl6i started2026-10-30T21:17:08.633UTC inUS-NC-1 atUSD0.89/hour (provider rate). Controller, bootstrap and validation exits are0. Final archive integrity, pod deletion and a fresh complete provider inventory confirmed cleanup.

Command: bash research/explicit_relations_e2e_20261030/run-arity.sh. Seeds101/202; tau=2ceil(r^(1/m))+2; rows2^32, work2^34, chunk2^20, columns16384, trials16. Solve caps180s at52bits and3600s at63bits; fresh replay60s, fixture180s, worker18000s/controller18300s. Finish every prescribed cell at the current rung, then stop larger rungs after any limit. The explicit threshold does not invoke the paper's sufficient L>=10m guarantee.

Source: e8e1c819df8901100db0254a849271130e6cca90. Prime-order public fixtures are reproduced in curves.txt, with p,r below2^63. Targets are coordinate lifts; no target logarithm is supplied to the reduction. The driver checks primality, curve membership and the unique Hasse-interval order witness. All28 Rust tests, host/device differential checks, Clippy, eight query variants under memcheck/racecheck/synccheck, service checks and maximum-cap headers passed. Every recovery passed a fresh CPU certificate replay and rejected scalar+1.

Cold solve includes per-target point generation, startup, validation, base construction, all relation attempts, elimination, extraction, internal replay, serialization and teardown. Build/preflight and shared curve generation are outside that interval. Separate audit totals include the fresh replay and negative control. CPU user/system time and measured peak VRAM were not captured. Periodic partial uploads add host I/O; they restore no solver state.

Protocol SHA256: b58d0dfe47bc039a6160ea7912374d37ff30f595e2daea262490beef58ac5422 Archive SHA256: d6c403d2eb5ccd4bc39a98432d38c70567efd9fd3f51445f47c4471305459237 The archive is under validation/ecbench-explicit-x/e2e-20261030-v19/results.tgz in the approved artifact bucket. Raw records were pushed in e1db39a661f3 before this report. Schema and file hashes pass. The strict native audit passes exact coverage and nine negative controls. The canonical rectangular checker retains four findings:16 assumed rows versus12, b/x each cover one rather than two sizes, and arm instance sets differ. canonical-check.log preserves these. Raw telemetry whitespace is preserved. A missing newer-version branch in the post-run audit's negative-control generator was repaired before audit passed.

Next: reduce factor-base sizes to fit24GB, with an explicit bounded target-query budget to allow complete relation collection from smaller bases.